SpaceNova

COLMAP-Based Indoor Reconstruction and Browser Roaming

Team: TechNova | **Competition:** Namma Space, Techfest IIT Bombay

Web viewer: https://space-nova-three.vercel.app
Source repository: https://github.com/Smita-142/SpaceNova

Abstract

SpaceNova is a web-based prototype for turning smartphone photos and video of an indoor environment into a browser-viewable 3D reconstruction. It combines a React interface and Three.js viewer with a FastAPI backend, OpenCV preprocessing, and COLMAP structure-from-motion (SfM). Users create a project, upload images or video, extract video frames, and run reconstruction. The resulting sparse point cloud can be inspected with orbit controls or explored in first person using mouse-look and WASD movement.

System architecture

The system separates the interactive client from the compute and storage backend. The React/Vite frontend is deployed on Vercel. It calls a FastAPI service deployed on Render for project management, media upload, frame extraction, reconstruction status, and model delivery. The backend stores uploads and reconstruction artifacts on persistent storage. OpenCV extracts video frames, while COLMAP performs feature extraction, image matching, and sparse mapping. The viewer loads the exported PLY point cloud; it can also load GLB and GLTF models.

Reconstruction pipeline and technical approach

The capture workflow targets sharp images with consistent lighting and substantial overlap between successive viewpoints. Video input is sampled at one frame per second by default. Before COLMAP processing, oversized staged images are downscaled to a maximum dimension of 2000 pixels to reduce CPU and memory costs; original uploaded media is retained. The CPU-compatible pipeline then extracts image features, exhaustively matches image pairs, and runs the COLMAP mapper to estimate camera poses and a sparse 3D structure. The sparse model is exported as a PLY point cloud for browser viewing.

Reconstruction runs asynchronously on the backend, allowing the client to poll job status and read progress logs while processing continues. Project media and reconstruction outputs are stored separately from the frontend so large uploads and generated models do not depend on a browser session.

Browser visualization

The web viewer provides orbit-based inspection and first-person roaming. In first-person mode, users click the scene to capture the mouse, use mouse movement to look around, and use WASD to move; Escape releases the pointer. Imported GLB/GLTF assets and COLMAP PLY outputs can be opened in the same viewer.

Evaluation considerations and limitations

The prototype demonstrates an end-to-end workflow from capture to online roaming. Reconstruction quality depends on image sharpness, lighting, texture, coverage, and overlap; reflective, transparent, moving, or textureless surfaces can reduce feature matching quality. The current COLMAP output is a sparse point cloud, not a dense, textured mesh. Processing runs on CPU, and large captures can take several minutes depending on image count and backend resources. These limits are surfaced so that the demo represents the current system accurately.

Reproducibility

The repository includes a compact, shareable sample capture and instructions for uploading it, running frame/reconstruction processing, and opening the result in the viewer. See the sample-data guide, capture SOP, and system architecture. The live viewer is available at space-nova-three.vercel.app; the source is at github.com/Smita-142/SpaceNova.

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